Economics After Cognitive Scarcity: The Permanent-Remainder Assumption and What Would Have to Be True for It to Hold

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Abstract

Arguments about what artificial intelligence will do to work keep one thing fixed while everything else moves: the assumption that whatever machines cannot yet do is a durable human province. This paper calls that the permanent-remainder assumption and computes what it would take to be true. Under a counterfactual in which every attainable cognitive service is free and universally available, three benchmarks are solved. An exact two-good equilibrium with identical preferences and identical free cognition yields consumption ratios of 81 to 1, 9 to 1 or 1 to 1 between the same two groups depending only on ownership and pooling, so equal intelligence does not narrow the distribution at all. A task model then separates two propositions the usual argument runs together: the set of tasks people still hold can be nonempty at every date while their share of tasks converges to $\delta/(a+\delta)$, where δ is task turnover and a the rate at which capability overtakes a new task. That limit does not depend on how fast new work appears, and a positive limit requires tasks needing something other than reasoning. Finally, the argument that automation must leave a human fallback holds across 84.85 percent of the plane of machine capability and failure independence and fails in a computable region, while past 50 percent automation of routine incidents an out-of-practice responder falls below plain containment. The paper's limits are stated: these are benchmarks with no prices, no investment and no empirical calibration.

The planner who was right

Imagine an official in 1900 asked to plan for a much larger New York. They know the traffic, the stabling, the feed and the street cleaning, and they produce a careful and correct estimate of how much manure the city will have to move. The arithmetic is sound. The answer is useless, because the horse was a contingent arrangement and the planner treated it as the structure of the problem.

The equivalent error in the present debate is not underestimating how capable machines will become. Someone can grant extraordinary capability and still describe the future as the present with different staffing: machines take the routine, people take the exceptions; machines produce answers, people check them; old occupations go, new occupations arrive. Each of these keeps a fixed object, the human role, and moves it to a new position inside an otherwise unchanged arrangement.

Economics has a name for the general hazard. The Lucas critique warned against treating estimated behavioural relations as invariant when a policy changes the environment those relations came from (Lucas 1976). The present case is not that argument and cannot borrow its authority: what is changing is technological capability and the organisational possibilities that follow, not a policy rule. The analogy is only that a relation can be correct inside a world and silent about the transition that ends it.

It also has to be said plainly that the strong version of the complaint, that nobody imagined any of this, is false. Simon wrote in 1960 that machines would within twenty years be capable of doing any work a person can do (Simon 1960). Marschak was pricing the cost of thinking as an economic input (Marschak 1968). Nilsson set out artificial intelligence, employment and income as a connected problem (Nilsson 1985). Romer built his growth theory on the distinction between a non-rival design and a person's rival capacity to execute it (Romer 1990), and Meade had already tied machine production to ownership and social dividends (Meade 1993). Contemporary work models automation of production and of research directly (Trammell and Korinek 2026; Growiec et al. 2023). The interesting question is which inferences survive when the premise is applied consistently, including to the activities used to preserve the familiar arrangement. Priority is a separate and less useful contest.

So the counterfactual is stated strictly. Every attainable cognitive service, at every quality, is free to develop, to copy and to execute, and everyone and every institution has unrestricted access. That includes reasoning about science, engineering, contracts, management, verification, persuasion, organisational design and public policy. It includes finding the errors in a plan and designing a better way to coordinate. It does not include omniscience, because inference cannot manufacture observations that were never made; it does not include free embodiment, because a perfect controller is not a machine; it does not include agreement, because people who understand each other perfectly can still want incompatible things; and it does not include authority, because knowing how to do something is not permission to do it. Those four exclusions are the only places a surviving human role can come from without contradicting the premise, and the rest of this paper is about what follows from that.

Identical intelligence, eighty-one to one

Take the simplest economy that can carry the question. There are 100 people, a general consumption good and a housing service, both in fixed net supply because their physical production is already automated. Preferences are identical, $\log c + \eta \log h$, and free cognition enters everyone's utility as the same additive constant, so it cannot affect how the two scarce goods are divided. Person i owns a share θ_i of every productive asset, and a fraction τ of asset income is pooled and shared equally. The competitive equilibrium is immediate: consumption and housing shares are both $s_i = (1-\tau)\theta_i + \tau/N$, and the relative price of housing is $\eta Q/H$.

Now let ten people own 90 percent of the assets and the remaining ninety own the rest, with both goods in supply 100. With no pooling, each of the ten consumes 9.000 and each of the ninety consumes 0.111, a ratio of 81 to 1. Pooling half of asset income gives 5.000 against 0.556, a ratio of 9 to 1. Pooling all of it gives 1.000 each. Intelligence, preferences, technology and total physical supplies are identical in all three rows.

This is a textbook construction, offered here for what it rules out rather than for itself. For any strictly positive vector of consumption shares summing to one there is an ownership vector that supports it, which was checked here against 200 randomly drawn target distributions with exact agreement. Universal cognitive equality therefore does not narrow the distribution of consumption at all: it is consistent with every distribution. Anyone who expects free intelligence to equalise outcomes owes an argument about ownership, and anyone who expects it to concentrate them owes the same argument in the other direction.

The second row of the experiment is quieter and matters more for how such a world would feel. Multiply the consumption good tenfold and change nothing else. Everyone's consumption of it rises tenfold. Housing quantities per person do not move at all, at 9.000 and 0.111 exactly as before; what changes is the relative price of housing, from 1.0 to 10.0. General material abundance and access to a particular constrained good are different things, and the second can be untouched by the first.

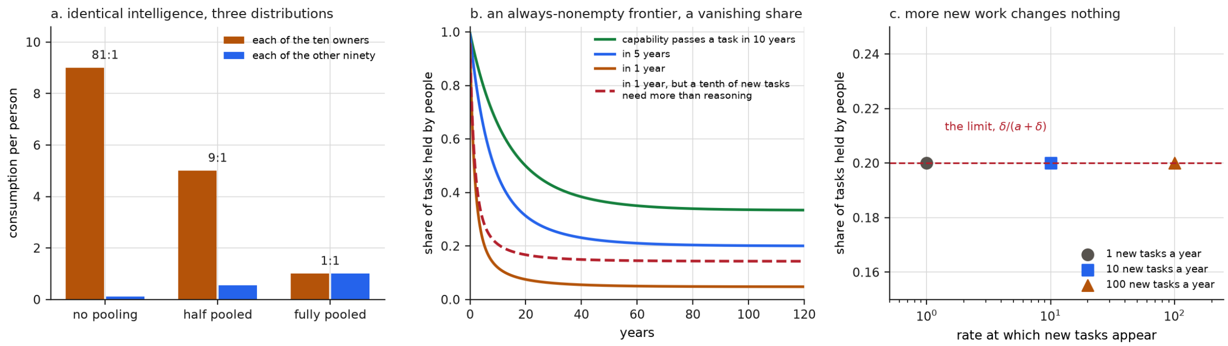

Figure 1: The allocation benchmark and the task frontier. (a) Consumption per person in each group under three pooling rules, with intelligence, preferences and physical supplies identical throughout. (b) The share of tasks held by people over time, for three rates at which capability overtakes a newly created task, and for the case where a tenth of new tasks require something other than reasoning. (c) The same limit under arrival rates spanning a hundredfold range.

The permanent-remainder assumption

The assumption is that whatever machines cannot yet do constitutes a durable human domain. It travels in several forms: machines take the routine and people the exceptions; machines answer and people verify; old jobs go and new jobs come. Each is a candidate hypothesis and none is absurd. What they share is a move from a statement about every date to a statement about some enduring thing, and that move is invalid. That at every date there exists a task where people have the advantage does not give that there exists a task where people have the advantage at every date.

Make it concrete. Tasks carry capability requirements. Machine capability rises, and a task is held by people while capability is below its requirement. A task created above the frontier is overtaken at rate a; every task becomes obsolete at rate δ; new tasks appear at rate ν. Then the share of tasks held by people converges to

$$\frac{\delta}{a+\delta},$$

which the simulation reproduces to within 5.50×10^{-5} across a grid of turnover and overtaking rates. Three things about this limit are worth separating.

The frontier is never empty. In every run the stock of human-held tasks is strictly positive at every date, because new tasks keep arriving above the frontier. The optimistic proposition is true, and it is compatible with the human share of tasks being 0.048 when capability overtakes a new task in a year and turnover is 5 percent a year. Both statements hold at once, which is precisely what the informal argument denies.

The arrival rate cancels. Varying ν from 1 to 100 new tasks a year changes the limiting share by 0.000000. This is the sharpest thing in the paper and the least intuitive. It is not that new work fails to appear; it is that new work and old work accumulate in the same proportion, so the share of it that people still hold is set by the race between turnover and capability, and not at all by how much new work there is. An argument that rests on the volume of new tasks is answering a question that does not bear on the quantity it claims to protect.

A positive limit requires something other than reasoning. Give a share of new tasks a requirement that capability never satisfies, which under the stated premise means physical presence, an unavailable observation, consent, or legitimate authority. With a tenth of new tasks of that kind the limiting share rises from 0.048 to 0.143. That is where a durable human role can come from, and it is the only place the premise leaves. The word "permanent" in the assumption is doing work that the atom, not the frontier, would have to supply.

None of this settles the income share, which is a further question and a harder one. Under the premise a human service that is a perfect substitute for a free cognitive service cannot command a positive competitive price for its cognitive content. So the share of tasks held by people is an upper bound on the share of income earned from them, and it binds only for the tasks whose requirement is not cognitive. Acemoglu and Restrepo model new tasks in which labour has a comparative advantage and are explicit that their framework also contains a regime in which all tasks are automated (Acemoglu and Restrepo 2018); Autor's account of why so many jobs remain is a careful argument about complementarity under conditions he states (Autor 2015). The criticism here is not of either. It is of the passage from a conditional result to an unconditional reassurance, which happens when the premises stop being repeated.

Where the fallback argument is right, and where it stops

A concrete case makes the difference between a good argument and an over-extension visible. Automated incident response now handles routine failures in production software, and a familiar worry follows: responders lose the practice that routine incidents used to provide, and are then less prepared for the hard ones that automation cannot handle.

The worry has a serious pedigree and it is substantially correct. Bainbridge's analysis of automation is built around systems in which a person is expected to take over in abnormal conditions, and observes that automation removes exactly the routine practice that keeps such a person capable (Bainbridge 1983). The difficulty has a name and a literature of its own, the out-of-the-loop performance problem, which studies how automation degrades an operator's readiness to resume control (Endsley and Kiris 1995). Where a person is a necessary recovery component, degrading that person's readiness degrades the system.

The over-extension is the step from there to the claim that a person must remain the recovery component. What matters when the primary system fails is a comparison of conditional success probabilities, and a fallback's value depends on how much of the primary's failure it shares. Model a practised responder with unconditional success 0.72 who also fails on a tenth of the primary's failures, a second automated system with capability κ that shares a fraction ρ of them, and a safe-state procedure that contains without diagnosing, succeeding 0.55 of the time.

Over the plane of capability and shared failure, the person is the right fallback in 84.85 percent of cells. That is a real vindication of the intuition and it should be said first. The person is not the right fallback everywhere: a second automated system wins wherever its independence is high enough, and the requirement is computable. It must share at most 0.341 of the primary's failures to beat the person at any capability at or below one. So the argument that automation needs a fallback is sound, and the argument that the fallback must be human is a claim about failure independence dressed as a claim about biology. Turing catalogued the general form of the move in 1950, granting demonstrated capabilities while asserting some further activity machines will never perform (Turing 1950).

The second computed result cuts the other way and lands where Bainbridge herself did. As routine incidents are automated, the responder's conditional success falls. It crosses the safe-state procedure at 50 percent automation of routine incidents: past that point an out-of-practice person, in this model, is worse than putting the system into a contained state and not diagnosing at all. Bainbridge recommends automatic shutdown where it is appropriate, and this is the arithmetic of that recommendation. The deskilling concern is not a reason to keep people in the loop. It is a reason to decide, deliberately, whether they are in it.

The third result concerns the evidence. Suppose mean time to resolve rises among the incidents people still handle. That is exactly what automation of the easy cases produces with competence held fixed, because the remaining sample is harder. With incident difficulty drawn from a gamma distribution and resolution time an exact linear function of difficulty, automating 75 percent of routine incidents raises the mean human resolution time by a factor of 1.88 with no change in competence whatsoever. A cohort whose competence degrades on an unchanged case mix shows a factor of 1.36. Combining both, selection accounts for 71 percent of the rise. An observed increase in resolution time is therefore not evidence of deskilling until the case mix is controlled, and the prediction that resolution times will rise is one both hypotheses make.

What free cognition does not dissolve

The counterfactual removes a constraint. It does not remove all of them, and the ones that survive are worth naming because they are what a serious argument about this world would be about.

Some impossibilities are not failures of reasoning. Myerson and Satterthwaite showed that under independent private valuations with overlapping supports, incentive compatibility, voluntary

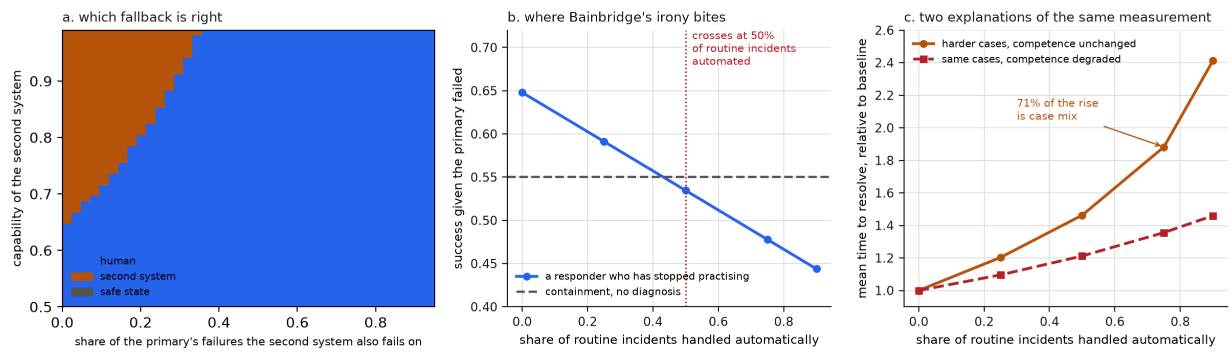

Figure 2: The fallback. (a) The best fallback over machine capability and the share of the primary's failures a second system also fails on; the safe-state procedure wins nowhere here, which is why panel (b) matters. (b) A responder's conditional success as routine practice is withdrawn, against plain containment. (c) Mean human resolution time under two hypotheses: harder remaining cases with competence unchanged, and unchanged cases with competence degraded.

participation and no outside subsidy, bilateral trade cannot generally be ex post efficient (Myerson and Satterthwaite 1983). No amount of computation repairs that; changing information access or permitting a subsidy does. Each institutional friction needs its own analysis rather than a blanket verdict in either direction.

Some mechanisms stop working at exactly zero rather than degrading smoothly. In a ratio contest for a fixed prize, effort at unit cost ε has a symmetric equilibrium whose aggregate expenditure is independent of ε (Tullock 2001). At $\varepsilon = 0$ there is no finite-effort equilibrium at all: any participant improves its odds by deploying more, forever. The correct statement is that the allocation rule has ceased to have the equilibrium anyone was extrapolating, and something else has to allocate. The old expenditure level is not still being paid in some hidden currency. Counting agents will not help, since a million identities under one principal are not a million claimants, which is the false-name problem in a new setting (Yokoo et al. 2004).

And the objective remains political. Free analysis can compare means to a given end; it does not select the end. Sen's distinction between democracy's instrumental role and its constructive one is the relevant one, since public deliberation is partly how a society forms the objectives it then pursues (Sen 1999). Ostrom's polycentric alternative to a single optimising authority is a live option in this world rather than a compromise forced by scarce expertise, because the cognitive cost of running many overlapping institutions has gone to zero along with everything else (Ostrom 2010). The best forecast is not the legitimate constitution.

The general discipline is one rule applied everywhere. Whenever the analysis introduces a bottleneck, ask whether it consists of cognitive work the premise has already made free. If it does, explain why it survives. If it does not, say which of the four exclusions it is. That rule is what stops the argument from quietly restoring expensive intelligence whenever the existing arrangement needs saving, and it is a rule this paper's own models have to obey too.

What is not being claimed

The models are benchmarks. Nothing in them forecasts anything. The allocation equilibrium holds supplies fixed and says nothing about investment; the task model has no prices in it and therefore bounds an income share rather than computing one; the fallback model prices a single incident class with stated success probabilities and no correlated cascades. None of the numbers is an estimate of any real economy, and the units in the task model are tasks rather than dollars for that reason.

Nor does the paper claim that anyone failed to imagine machine-dominated production. That claim is false and would be a distraction; the cited predecessors state their premises clearly, and the frameworks that contain a full-automation regime say so (Acemoglu and Restrepo 2018). The target is narrower and, I think, worth hitting: an inference that survives by ceasing to repeat its conditions. The permanent-remainder assumption is almost never asserted. It is relied upon.

What follows for anyone making the argument is a short list. If the claim is that people will keep a share of output, say whether it rests on an always-nonempty frontier, which the model shows is compatible with a vanishing share, or on tasks with a non-cognitive requirement, which is the only thing that produces a positive limit. If the claim is that new work will appear, note that its volume does not enter the limit. If the claim is that a person must remain the fallback, state the failure independence that would make it true, since the region where it holds is computable and it is not the whole plane. And if the claim is that observed resolution times prove deskilling, control for the case mix first, because most of the effect is there.

The last point is the one I would keep. A society that defends people's worth by insisting they will always beat machines at something economically useful has made that worth depend on a contest it never needed to enter, and has agreed to lose the argument the moment the contest goes the other way. The allocation benchmark makes the alternative arithmetic rather than sentiment: with identical intelligence and identical technology, the distribution of consumption ran from 81 to 1 down to 1 to 1 on the strength of enforceable claims alone. What people can do turned out not to determine what they get. What they are owed was never going to be settled by finding them a residual job description.

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